Neural Prediction of Bilingual Code-Switching Through Eye-tracking and Machine Learning Classification

Project Description

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GitHub: https://github.com/selinah66/NeurotechUSC-Bilingual-Code-Switching/tree/main

Background:

Bilingual language processing has been widely studied, revealing distinct eye movement patterns in bilinguals reading in their second language (L2). Bilinguals typically show longer reading times, more fixations, and shorter saccades compared to monolinguals, reflecting increased cognitive load (Mou et al., 2024). The Weaker Links Hypothesis suggests this occurs due to divided cognitive resources between languages (Cop et al., 2015), though higher L2 proficiency can reduce these effects (Quiñonez-Beltran et al., 2024). Additionally, language switching incurs cognitive costs, with more pronounced switching behavior among frequent code-switchers (Gollan & Ferreira, 2009). Eye-tracking technology provides valuable metrics, such as gaze duration and fixation count, which can reveal these cognitive processes. This research aims to explore how these eye movement patterns, particularly in code-switched sentences, can predict bilingual language proficiency, offering a deeper understanding of bilingual cognitive control during reading.

Research Question / Objective:

The objective of this project is to examine the relationship between language switching (code-switching) and visual attention in bilingual individuals. Specifically, we aim to leverage eye-tracking data to predict bilingual participants' proficiency in a second language. Our approach will analyze eye-tracking metrics from bilinguals reading code-switched Chinese-English sentences, utilizing a machine learning (ML)-based Random Forests classification model.

We will begin by training our model using an open-source eye-tracking dataset, followed by validation and testing on newly collected data from bilingual participants. The validation process will account for variables such as emotional salience and other contextual factors influencing attention during language switching. The ultimate goal is to develop a predictive model capable of assessing bilingual proficiency, offering new insights into cognitive processing in bilinguals and informing language learning strategies.

Scientific and Social Impact

This project investigates the cognitive mechanisms underlying bilingual language switching by analyzing visual attention patterns through eye-tracking. Understanding how bilinguals switch between languages provides valuable insights into brain function and cognitive control. By

combining eye-tracking technology with machine learning, we aim to develop a predictive model linking eye movement patterns to language proficiency, offering new insights into bilingual cognitive processes.

The broader significance of this research includes reshaping how we approach language acquisition and proficiency assessment. Predicting proficiency using real-time data opens the door to more accurate, efficient language assessment tools for educational and clinical settings. This project could also contribute to AI systems that better understand bilingual language use, advancing machine translation, natural language processing, and human-computer interaction.

Moreover, the findings could influence bilingual education, where understanding cognitive load and language-switching patterns can inform personalized teaching strategies. Ultimately, this project bridges cognitive science, linguistics, neuroscience, and technology, making it an impactful investment for both theoretical and practical applications.

Methodology:

This study consists of two parts: leveraging existing eye-tracking data to develop an initial Random Forests classifier model and collecting our own eye-tracking data to refine the model.

1. Existing Data

We will start by using an open-source eye-tracking dataset from Chinese-English bilinguals, with 80 participants categorized by their English proficiency (low and high). The dataset includes key eye-tracking metrics such as Regression Path Duration, First Fixation Duration, Gaze Duration, and Regression In/Out Counts, providing insights into visual attention during code-switched sentence reading. This dataset will serve as the foundation for training our machine learning model.

2. Data Preprocessing and Feature Engineering

Data cleaning will involve filtering missing values, removing outliers, and performing quality checks. We will then use feature engineering to select the most relevant variables through correlation analysis and linear regression, ensuring the model focuses on the strongest predictors of bilingual proficiency.

3. Model Training

We will split the data into 70% training and 30% testing subsets. The model will be trained using the Random Forest classification technique, predicting language proficiency based on eye-tracking metrics. Model performance will be evaluated by classification accuracy.

4. Future Data Collection

To refine the model, we plan to collect our own eye-tracking data from bilingual participants and design our own linguistic stimuli for participants to use while collecting the data. This will allow us to tailor the data to our research needs, address gaps in the existing dataset, and improve the model's accuracy and generalizability.

5. Expected Outcomes

The outcome will be a machine learning model capable of predicting bilingual proficiency based on eye-tracking data, contributing to understanding cognitive processes in bilinguals and potentially improving language proficiency assessment tools in both educational and clinical contexts.

References:

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